

You are going to calculate the lattice constant for <Crystal-structure> <Species> through DFT. The experimental value is xxx; use this to create the initial structure.



User

Supervisor



Plan
Generated

Structure
Generation



DFT Agent

Bulk Structure
Generation

Convergence
Test Setup



DFT Agent

Pseudopotential
Look Up

Template
Script
Creation

Convergence
Test
Preparation

Convergence
Job Submission



HPC Agent

Resource
Allocation

Job
Submission

Production
Script
Generation



DFT Agent

Result
Extraction

Determine Optimal
Production Run Settings

EOS Script
Generation

Production
Job Submission



HPC Agent

Resource
Allocation

Job
Submission

Calculate Lattice
Constant



DFT Agent

Extract
Result

Lattice
Constant
Calculation

Answer: Lattice
constant for
<Species> with
<Crystal-structure>
crystal structure is
xxx